

CHARLOTTE, MI, USA

WMO: 720284

Lat: 42.574N

Lon: 84.811W

Elev: 268

StdP: 98.15

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 04877

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.3	-15.1	-22.2	0.5	-17.6	-18.9	0.7	-14.1	9.8	-4.2	8.9	-4.0	1.3	200	0.502

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.9	31.9	22.4	30.0	21.6	28.3	20.7	23.9	28.9	23.0	27.7	22.2	26.7	3.9	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.4	17.7	26.8	21.4	16.6	25.6	20.8	16.0	25.0	73.0	29.1	69.4	27.7	66.2	26.5	27.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.7	7.7	7.0	-23.1	33.7	3.8	1.8	-25.8	35.0	-28.0	36.1	-30.1	37.1	-32.8	38.4		
			-23.3	25.8	3.7	1.0	-26.0	26.5	-28.2	27.0	-30.2	27.6	-32.9	28.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.1	-4.4	-3.9	1.7	8.4	14.6	19.6	21.5	20.4	17.0	10.4	4.0	-1.1
	DBStd	10.42	6.26	6.09	6.17	4.91	4.81	3.47	3.18	2.98	4.10	4.78	4.95	5.11
	HDD10.0	1798	447	391	268	88	12	0	0	0	2	55	190	346
	HDD18.3	3686	704	623	516	301	136	25	8	13	73	251	431	603
	CDD10.0	1461	1	1	11	39	155	288	356	323	211	67	9	1
	CDD18.3	307	0	0	1	1	21	63	106	78	33	4	0	0
	CDH23.3	2852	0	0	8	16	237	574	1015	664	299	38	1	0
	CDH26.7	791	0	0	1	1	54	166	324	163	78	4	0	0
Wind	WSAvg	2.8	3.3	3.3	3.2	3.5	2.8	2.4	2.1	1.9	2.0	2.8	3.1	3.3
Precipitation	PrecAvg	868	53	44	54	82	102	96	83	87	72	78	67	50
	PrecMax	1058	108	100	108	199	266	215	154	171	224	155	150	104
	PrecMin	664	14	18	12	15	38	18	26	16	19	6	9	18
	PrecStd	107	27	20	26	40	51	50	30	41	44	34	36	18
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.1	14.9	24.1	25.9	31.0	32.4	34.1	32.5	31.8	27.1	20.9	14.9
		MCWB	12.2	11.5	17.2	15.8	20.5	22.5	22.9	22.5	21.9	20.1	13.8	13.0
	2%	DB	9.3	10.2	18.0	22.6	27.9	30.6	32.1	30.1	28.9	24.1	16.7	11.1
		MCWB	7.7	8.1	13.1	14.1	19.4	21.0	23.2	22.0	21.3	17.5	11.7	9.3
	5%	DB	6.2	6.3	14.5	20.2	26.2	28.6	30.0	28.1	27.0	21.3	14.1	8.0
		MCWB	4.5	4.0	10.7	12.9	18.3	20.8	22.0	20.7	19.7	16.3	10.4	6.4
	10%	DB	3.7	3.8	11.3	17.5	23.7	27.0	28.0	27.2	24.8	18.9	12.1	5.2
		MCWB	2.6	2.1	8.0	11.7	16.9	19.5	20.8	20.3	18.8	15.1	9.1	3.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.2	12.5	17.2	17.8	22.2	24.0	25.7	24.8	23.8	21.3	15.2	13.4
		MCDB	13.0	13.6	23.7	23.2	28.6	29.6	31.3	29.5	29.0	25.5	17.9	14.7
	2%	WB	8.2	8.3	14.3	15.8	20.7	22.8	24.3	23.3	22.6	19.2	12.8	9.4
		MCDB	8.7	10.7	17.6	19.7	26.0	28.3	29.6	27.5	27.1	22.3	16.2	10.5
	5%	WB	4.9	4.4	11.6	14.0	19.6	21.8	23.2	22.4	21.3	17.3	11.0	6.4
		MCDB	6.2	6.2	14.1	18.9	24.4	26.9	28.1	26.5	24.7	19.9	13.5	7.7
	10%	WB	2.7	2.2	7.9	12.1	18.1	20.8	22.3	21.5	19.9	15.3	8.9	3.8
		MCDB	3.6	3.7	10.5	16.7	22.3	24.9	26.6	25.3	23.5	18.7	11.7	5.1
Mean Daily Temperature Range		MDBR	7.4	8.3	9.7	11.5	11.8	11.3	11.9	11.6	11.8	10.4	8.3	6.2
	5% DB	MCDBR	9.2	10.3	14.5	15.3	14.2	13.0	13.4	13.1	13.9	14.1	12.3	9.4
		MCWBR	8.0	8.7	9.6	8.7	7.2	5.9	5.9	6.1	6.8	8.4	8.5	8.2
		MCDBR	8.7	10.0	13.0	13.1	12.3	10.8	11.1	10.4	11.1	11.8	11.2	9.1
	5% WB	MCWBR	7.9	8.9	9.4	9.1	6.9	5.9	5.6	5.6	6.2	8.5	8.7	8.5
Clear-Sky Solar Irradiance	taub	0.297	0.358	0.341	0.387	0.419	0.440	0.428	0.426	0.384	0.346	0.315	0.298	
	taud	2.357	2.137	2.360	2.255	2.213	2.194	2.231	2.234	2.336	2.437	2.471	2.407	
	Ebn at Noon	840	822	901	880	857	837	843	832	844	838	817	807	
	Edh at Noon	81	120	109	131	140	143	137	132	111	89	73	71	
All-Sky Solar Radiation	RadAvg	1.42	2.29	3.57	4.59	5.35	6.04	6.14	5.26	4.24	2.70	1.67	1.14	
	RadStd	0.13	0.30	0.36	0.43	0.50	0.39	0.28	0.26	0.40	0.27	0.22	0.12	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (62 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page